The Component-Spectra-Parameterized Angular and Spectral Kernel-Driven Model: A Potential Solution for Global BRDF/Albedo Retrieval From Multisensor Satellite Data

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***Abstract*— The angular and spectral kernel-driven (ASK) model distinguishes soil and vegetation spectral features by the component spectra and is a promising model which combines multisensor data for inversion. However, its global application is limited by the component spectra. This article proposes parameterization of the ASK component spectra of soil and leaf from global spectra libraries as ANGERS, GOSPEL, LOPEX, and USGS. A statistical ratio (γ) of various leaf to soil spectra is used to capture their spectral differences and variations, with mean (m) + u (0, ±0.5, ±1) standard deviations (σ) [i.e., γ ($m + u\sigma$)]. Optimization inversion is applied to determine the ratio candidates $\gamma(m + u\sigma)$, allowing more tolerance for spectral uncertainty, which releases the semiempirical nature of the kernel-driven model. Simulation data analysis proves its feasibility and good capture of vegetation-soil spectral differences. The model's bidirectional reflectance factor (BRF) fitting error [root-mean-square error (RMSE)] of 0.0245 is slightly larger than the true component spectra of 0.0178, and albedo RMSE is 0.0116 in Black Sky Albedo and 0.0182 in White Sky Albedo. The result also shows its good robustness to the noises, where the level up to 20% noise conducts a 0.0277 error in BRF fitting and an ignorable influence in albedo. The synergistic-retrieved albedo from multisensor satellite data consists of *in situ* measurements with an RMSE of 0.0171, compared to 0.0131 from true component spectra retrievals. The new parameterization sacrifices some accuracy, but it is simple and operational for global retrieval with a satisfactory precision.**

***Index Terms*— Albedo, angular and spectral kernel-driven (ASK) model, bidirectional reflectance function (BRDF), multisensor satellite data.**

I. Introduction

BIDIRECTIONAL reflectance function (BRDF) describes the intrinsic and distinctive property of the land surface reflectance anisotropy, and thus plays an important role in optical quantitative remote sensing [1], [2]. BRDF is often used to infer the land surface information such as leaf area index (LAI) [3], vegetation structural parameters [4], [5], vegetation biophysical parameters [6], aerodynamic surface roughness [7], and snow grain size and pollution amount [8]. Especially, it allows a more accurate estimation of land surface albedo, which is a key parameter in energy budget and global change [9], [10]. Another utilization of BRDF is to correct angular effects, which is essential in the classification and dynamic monitoring [11].

Since 2000, global BRDF/Albedo products are derived from various satellite data with temporal resolution ranging from daily to ten plus days, such as the Moderate Resolution Imaging Spectroradiometer (MODIS) BRDF and albedo product V006 (MCD43 V6) [12], Polarization and Directionality of the Earth's Reflectances (POLDER) albedo [13], Multiangle Imaging SpectroRadiometer (MISR) albedo [14], CYCLOPES [15], Globalbedo [16], and Geoland 1 and 2 [17], [18]. These products have routinely retrieved BRDF from multiangular observations in clear-sky conditions. Taking the daily product MCD43 V6 as an example, BRDF is inversed from the 16-day-accumulated directional reflectance surrounding the central day of interest [12]. These multiangular observations are usually accumulated from multiday revisits of the same target from a single sensor (like POLDER BRDF product) or a single kind of sensor with the same sensor configuration (like MCD43 BRDF product). In addition, full

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inversion is often limited by insufficient angles mainly due to cloud contamination as up to 55% cloud fraction over land derived from MODIS cloud mask [20] and approximately 80% at global scale [21], and sometimes due to mislabels as cloud for bright pixels like desert or snow in cloud detection [22]. Even for the MISR, imaging the Earth in nine different view directions costs 16 days for global cover due to the narrow field of view, thus facing more challenges while obtaining global multiangular data.

The deficient angular sampling and coarse temporal resolution can be conquered by the synergy retrieval of combining multisensor data [23]–[25]. What is more, a long-term BRDF/Albedo product with temporal-and-spatial continuity can be achieved by integrating multisensor data, which is important for decadal-climate research [26]. However, challenges for combining multisensor data for synergistic retrieval include variations in spectral and spatial responses, differences in calibration, geometry registration, and especially developing a feasible model for global complex scenes [21], [25], [27], [28].

The angular and spectral kernel-driven model (ASK model), which will be discussed in Section II, decouples the radiometric and structural variables by the component spectra accounting for the variation in spectral bidirectional reflectance factor (BRF) with sensor bands [29]. Thus, ASK shows great potential in combining multisensor data to derive BRDF/Albedo with high accuracy in a shorter time period. However, the retrieval accuracy depends on the selection of component spectra, and it is not operational to assign the exact component spectra to each retrieval. Therefore, it is not applicable to global production [30].

This article proposes a parameterized ASK model. The component spectra are statistically parameterized by utilizing amounts of spectra from globally collected spectral libraries. The parameterized ASK is expected to be a compromise; whereas with some precision loss and still assuring acceptable accuracy, it can realize global BRDF retrieval fast and operationally. This article is organized as follows: Section II presents a brief overview of the ASK and then Section III proposes the parameterized ASK. Section IV describes the datasets used in this article. Section V derives the parameterized spectra and investigates its performance by using simulated BRF datasets. The satellite data application and validation are carried out in Section VI. Finally, the conclusions and perspectives are summarized in Section VII.

II. Overview of ASK Model

Developed from the kernel-driven model, ASK decouples the radiometric and structural parameters in the original coefficients and inserts the radiometric variables into the kernel functions with band-specific component spectra characterizing the sensors' spectral responses, which is shown as follows:

$$
\begin{aligned}
R(\Theta,\lambda) &= c_1k_1(\lambda)+c_2k_2(\lambda)+c_3k_3(\Theta,\lambda)\\
&\quad +c_4k_4(\Theta,\lambda)+c_5k_5(\Theta,\lambda)\\
k_1(\lambda) &= \frac{1}{\pi}\rho_g(m_0,\lambda)\\
k_2(\lambda) &= -\frac{1}{\pi}A_w(\lambda)\cdot\rho_g(m_0,\lambda)\\
k_3(\Theta,\lambda) &= \frac{2}{3\pi}\rho_c(\lambda)\cdot\left(k^c_{\text{geo}}(\Theta)+1\right)+\frac{1}{\pi}k^g_{\text{geo}}(\Theta)\cdot\rho_g(m_0,\lambda)\\
k_4(\Theta,\lambda) &= -\frac{1}{\pi}k^g_{\text{geo}}(\Theta)\cdot A_w(\lambda)\cdot\rho_g(m_0,\lambda)\\
k_5(\Theta,\lambda) &= \frac{2}{3\pi^2}\rho_c(\lambda)\cdot k^{\rho}_{vol}(\Theta)+\frac{2}{3\pi^2}\tau_c(\lambda)\cdot k^{\tau}_{vol}(\Theta)\\
&\quad +\frac{1}{3\pi}\rho_c(\lambda)+\frac{1-\sqrt{1-\omega(\lambda)}}{\pi(1+2\cos(\theta_i)\sqrt{1-\omega(\lambda)})}\\
c_1 &= a_1\cdot(1-n\pi r^2)+a_2\cdot\exp(-bF)\\
c_2 &= a_1\cdot(1-n\pi r^2)\cdot(m_1-m_0)\\
&\quad +a_2\cdot\exp(-bF)\cdot(m_2-m_0)\\
c_3 &= a_1\cdot n\pi r^2\\
c_4 &= a_1\cdot n\pi r^2\cdot(m_1-m_0)\\
c_5 &= a_2\cdot(1-\exp(-bF))
\end{aligned}
\tag{1}
$$

where R is BRDF (R can also be expressed as BRF if multiplying π in the equation's right side), Θ is the illumination and observation geometry, λ is the spectral band. k_1, k_3, and k_5 are the isotropic, geometric, and volumetric kernels, respectively, k_2 and k_4 are kernels accounting for the soil moisture. $k^g_{\text{geo}}k^g_{\text{geo}}$ and $k^c_{\text{geo}}k^c_{\text{geo}}$ are the ground and canopy parts in the Li-sparse kernel, and $k^{\rho}_{\text{vol}}k^{\rho}_{\text{vol}}$ and $k^{\tau}_{\text{vol}}k^{\tau}_{\text{vol}}$ are the reflectance and transmittance terms in the Ross-thick kernel, respectively. c_1–c_5 denote the five spectral-independent kernel coefficients. Leaf reflectance $\rho_c(\lambda)$, leaf transmittance $\tau_c(\lambda)$, soil reflectance $\rho_g(\lambda)$, soil moisture attenuating factor $\text{A}_\omega(\lambda)$, and single scattering albedo $\omega(\lambda)$ are band-specific component spectra.

For specific sensor m and band n, band-specific component spectra CS_{mn} has quantified the radiometric features of the land surface and the sensors' spectral responses. It is integrated from the band spectral response function $B_{mn}(\Lambda)$ with wavelength-dependent priori-knowledge $r(\Lambda)$ as shown in the following equation, where Λ is the wavelength and Λ_1 to Λ_2 is wavelength range of the spectral band:

$$
\text{CS}_{mn}=\frac{\int_{\Lambda_1}^{\Lambda_2} r(\Lambda)\cdot B_{mn}(\Lambda)d\Lambda}{\int_{\Lambda_1}^{\Lambda_2} B_{mn}(\Lambda)d\Lambda}. \tag{2}
$$

Among these component spectra, the soil moisture attenuating factor $\text{A}\omega(\Lambda)$ is a fixed spectrum independent of the soil targets accounting for the water variance effects in the soil reflectance (an exponential relationship of soil moisture with $\text{A}\omega$) [29], which is measured from laboratory experiments [31]. The leaf single-scattering albedo equals to the sum of leaf reflectance and transmittance. Hence, there are three remaining variables, i.e., leaf reflectance, leaf transmittance, and soil reflectance. They should be collected beforehand according to the target's compositions (vegetation and soil) in the routine inversion of ASK.

The component spectra have brought significant effects in ASK performance [30]. One is the kernel shape variance (i.e., angular characteristics). The ASK geometric optical kernel (abbreviated as geokernel) with separated vegetation and soil reflectance has changed kernel shapes dramatically from the original geokernel. While a slight shape distortion

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happens in ASK volumetric scattering kernel (abbreviated as vol-kernel) which separates the leaf transmittance and leaf reflectance compared to the original vol-kernel. The other is the albedo retrieval uncertainty caused by inaccurate component spectra. When the component spectra uncertainty is only from soil, albedo is very close to the true estimation from correct component spectra, with an average absolute error of 0.005. Because uncertainty is only from leaf, it exhibits an average absolute error of 0.012. And, uncertainties from incorrect soil and leaf spectra are additives upto some degrees with an error of 0.017.

As seen from the aforementioned results, the spectral difference between vegetation and soil are strongly affecting angular features when the differences of leaf transmittance and leaf reflectance are ignorable. The retrieval accuracy depends more on leaf component spectra, while it is not so sensitive to the soil component. In addition, simplification of assigning leaf transmittance with leaf reflectance in ASK inversion obtains a satisfactory accuracy. It is reasonable to make an approximation of leaf transmittance by leaf reflectance in ASK. Therefore, the leaf reflectance and soil reflectance are the two key variables to be determined in ASK performance.

III. Parameterization Method of ASK Model

Kernel-driven models of vegetated land surfaces have well-captured the real scenarios' bidirectional reflectance features by linearly combining the surface scattering model and the volumetric scattering model [32]. The surface scattering term (geokernel) based on the geometric optical theory specifies a basic BRDF as the different viewed proportions of sunlit canopy and sunlit soil according to the observation geometries. This is significant in the sparse canopy simulated as obstructions or protrusions like shrubs and trees arranged on soil background. The volumetric scattering term (vol-kernel) based on the radiation transfer theory describes the bidirectional scattering in a turbid medium consisting of small plane facets or particles, e.g., leaves over a soil background. For both typical scatterings, the composition of vegetation and soil determines the BRDF, including not only their proportions in the field of view but also their spectral features. In the original geokernel, the sunlit canopy and soil are simplified with the same reflectance. Nevertheless, such two components show very different spectral features. A key point is to differentiate their spectral responses in the kernel-driven model. This has already been proven in the component spectra effects in the ASK performance.

Considering the ASK model's property, a ratio between the two components (i.e., the leaf reflectance and the soil reflectance) is of great convenience and feasibility to express their spectral differences. As a nature of the kernel-driven model, coefficients are semiempirically adapted to fit a given target. For the spectral variables, ASK can also adjust the kernel coefficients allowing some tolerance for component spectra. Consequently, compensation or cooperation happens between the soil and leaf spectral information. With such an advantage, a relative difference between them can satisfy application requirements instead of soil and leaf absolute spectral reflectance values. The ratio form is commonly used to express the information from two different variables like in vegetation indexes inventory. The leaf reflectance varies more with wavelength compared to the soil reflectance, taking, for example, the well-known red edge effect in the leaf spectrum. To avoid extreme values in ratio, the denominator should be steadier. Therefore, the ratio is designed as the leaf reflectance to the soil reflectance, which is shown as follows:

$$\gamma(\lambda) = \rho_c(\lambda)/\rho_g(\lambda). \tag{3}$$

As replacing leaf transmittance by leaf reflectance will make acceptable effects, we set $\rho_\tau(\lambda) = \rho_c(\lambda)$. Then, ASK is parameterized with the ratio and soil reflectance as follows:

$$\begin{aligned}
R(\Theta, \gamma(\lambda), \rho_g(\lambda)) &= c_1 k_1(\rho_g(\lambda)) + c_2 k_2(\rho_g(\lambda)) \\
&\quad + c_3 k_3(\Theta, \gamma(\lambda), \rho_g(\lambda)) \\
&\quad + c_4 k_4(\Theta, \rho_g(\lambda)) \\
&\quad + c_5 k_5(\Theta, \gamma(\lambda), \rho_g(\lambda)) \\
k_1(\rho_g(\lambda)) &= \frac{1}{\pi}\rho_g(\lambda) \\
k_2(\rho_g(\lambda)) &= -\frac{1}{\pi}A_w(\lambda)\cdot\rho_g(\lambda) \\
k_3(\Theta, \gamma(\lambda), \rho_g(\lambda)) &= \frac{2}{3\pi}\gamma(\lambda)\cdot\rho_g(\lambda)\cdot(k^c_{\text{geo}}(\Theta)+1) \\
&\quad + \frac{1}{\pi}k^g_{\text{geo}}(\Theta)\cdot\rho_g(\lambda) \\
k_4(\Theta, \rho_g(\lambda)) &= -\frac{1}{\pi}k^g_{\text{geo}}(\Theta)\cdot A_w(\lambda)\cdot\rho_g(\lambda) \\
k_5(\Theta, \gamma(\lambda), \rho_g(\lambda)) &= \frac{2}{3\pi^2}\gamma(\lambda)\cdot\rho_g(\lambda)\cdot(k^\rho_{vol}(\Theta)+k^\tau_{vol}(\Theta)) \\
&\quad + \frac{1}{3\pi}\gamma(\lambda)\cdot\rho_g(\lambda) \\
&\quad + \frac{1-\sqrt{1-2\gamma(\lambda)\cdot\rho_g(\lambda)}}{\pi(1+2\cos(\theta_i)\sqrt{1-2\gamma(\lambda)\cdot\rho_g(\lambda)})}.
\end{aligned} \tag{4}$$

IV. Data

To assign the ratio and soil reflectance in the parameterized ASK (4), global spectral libraries of leaf and soil are utilized. The parameterization performance is analyzed and evaluated using simulated BRF datasets. And as an application of the parameterized ASK, multi-sensor satellite data are employed to retrieve albedo, and *in situ* measurements are used to validate the retrieved albedo.

A. Spectral Libraries for Parameterization

The globally distributed soil and leaf spectra are involved to interpret the spectral complexity, covering visible to near-infrared range of 400 to 2500 nm.

Leaf spectra are collected from LOPEX, USGS, and GOSPEL covering 59 species, including Broadleaf (29), Pine leaf (3), Crops (15), Grass (9), and Shrubs (3), shown in Fig. 1. All these species are representative of the types that have been encountered and studied in published works, covering most of the biomes in the International Geosphere-Biosphere Programme (IGBP). The spectra in the Leaf Optical Properties Experiment 93 (LOPEX93) are measured by the Joint

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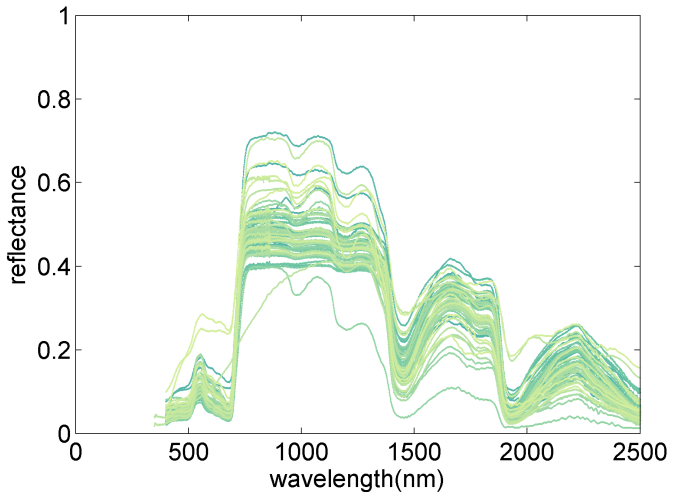


Fig. 1. Leaf spectra profiles of different species from three spectra libraries.

Research Centre in the laboratory with ancillary physicochemical parameters [33]. Vegetation library in USGS compiled spectra of trees, shrubs, grasses, flowers, leaves, and lichens and other microorganisms, including laboratory spectra of leaves from a single plant and field spectra of a single species, or of a plant in its entirety [34]. Ground Object SPEctral Library (GOSPEL) has compiled and measured the vegetation spectra from leaf to canopy scales in the field with ancillary physicochemical and structural information, which is built by the Institute of Remote Sensing and Digital Earth (RADI), Chinese Academy of Sciences (CAS) [35].

The soil spectral library from the International Centre for Research in Agroforestry (ICRAF) is employed here due to its global distribution and relatively complete ancillary information (especially the systemic soil classification). It consists of more than 3000 spectral profiles from 58 countries in Africa, Asia, Europe, North America, and South America, and associated attributes as the country, geographical coordinates, soil classification, and physical and chemical properties [36]. The systemic soil classification is with good representation on the global distribution and geography of soils according to the Food and Agriculture Organization (FAO) 1988. It is a two-level hierarchical system of 28 major soil units at the first level mainly differentiated by the primary pedogenetic process producing the characteristic soil features, and 153 units at the second level according to predominant secondary soil-forming process significantly affecting the primary soil features (www.fao.org/3/w8594e/w8594e03.htm). Compromising the complexity and feasibility in operation, the first-level classification is investigated here, which has been listed in Table I. And in total, 3188 soil spectra samples are selected for parameterization where spectra with incomplete information are eliminated. The remaining profiles contain 20 units covering most of the land surface according to the world soil units' map from the Harmonized World Soil Database [37].

B. Simulated BRF Data Sets for Evaluation

To avoid the unknown noises in real satellite data, simulated BRF datasets are used to evaluate the parameterization performance. The Sail model [38] and four-scale model [39] are used to simulate BRFs of continuous and discrete canopies, respectively. As this article concerns more about the spectral issues, the parameter setting in the simulation focuses more on the spectral variances than the structural ones. Each model has input 20 species of leaves and 40 randomly selected samples of soil (two in each soil unit). Five different LAIs are set to simulate from nearly bare scenes to well-vegetated scenes. There are 8000 ($2 \times 20 \times 40 \times 5$) sets of BRFs in total. The scenes' spectral reflectance is simulated under 32 actual MODIS observation geometries extracted from a pixel during a 16-day cycle both in Terra and Aqua, and then resampled with spectral response functions from MODIS optical bands to derive the MODIS-like BRFs, centered at 0.648, 0.858, 0.470, 0.555, 1.240, 1.640, and 2.13 μm, respectively. The main parameters for each model simulation are listed in Tables II and III, and some examples of the MODIS-like BRFs are shown in Fig. 2.

TABLE I
ABBREVIATION (ABBR.) FOR THE 28 MAJOR SOIL GROUPINGS OF FIRST-LEVEL CLASS IN THE FAO/UNESCO SOIL CLASSIFICATION SYSTEM

Abbr.	Soil unit	Abbr.	Soil unit	Abbr.	Soil unit
AC	Acrisols	GL	Gleysols	PL	Planosols
AL	**Alisols**	GR	Greyzems	**PT**	**Plinthosols**
AN	Andosols	**GY**	**Gypsisols**	PZ	Podzols
AT	**Anthrosols**	**HS**	**Histosols**	PD	Podzoluvisols
AR	Arenosols	KS	Kastanozems	RG	Regosols
CL	**Calcisols**	**LP**	**Leptosols**	SC	Solonchaks
CM	Cambisols	LX	Lixisols	SN	Solonetz
CH	Chernozems	LV	Luvisols	VR	Vertisols
FR	Ferralsols	**NT**	**Nitisols**		
FL	Fluvisols	PH	Phaeozems		

Note: the bolded text are the absent soil units in this study due to the data absence

TABLE II
SAIL MODEL PARAMETERS

Parameter	Values
Component spectra	Leaf and Soil Spectral libraries
Diffuse skylight factor	0.1
Leaf angle distribution	Custom
LAI	0.3, 1.0, 2.0,4.0,5.0
Illumination-view geometry	MODIS true observation geometry

C. Satellite Data and In Situ Data for Application and Validation

This article, as a part of ASK's series, is a further step for the operational retrieval in global satellite data. Thus, the same satellite data and *in situ* measurements in [30] are utilized to apply and validate the parameterization. The Heihe River Basin in the northwest of China is the study area, with relatively homogeneous geography and variable landscapes including barrens, desert, the Gobi desert, croplands,

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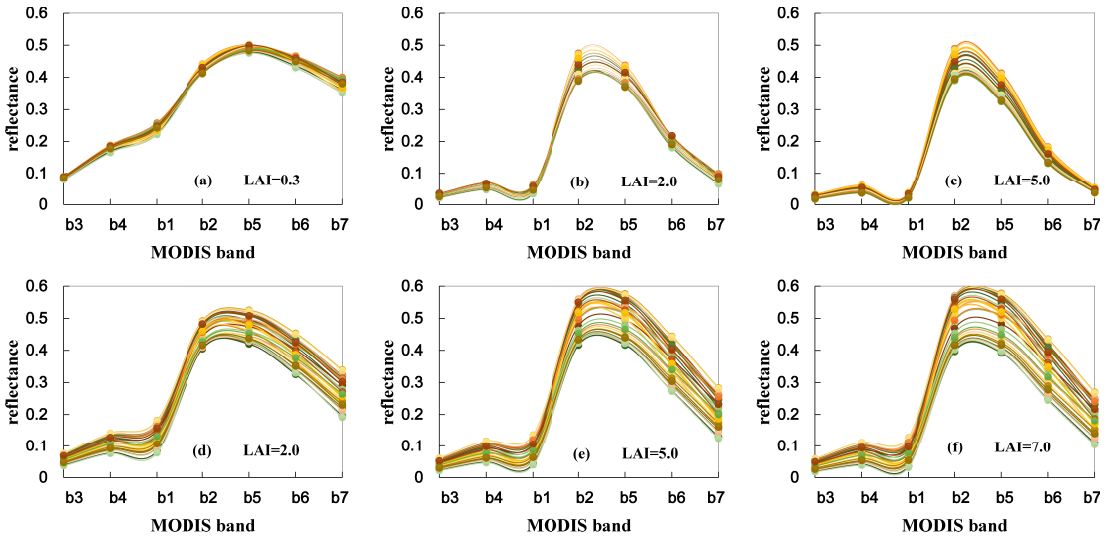


Fig. 2. Simulated MODIS BRFs of 32 observation geometries by SAIL and 4SCALE from low to high LAI with the same set of spectral and structural parameters. The first row (a)–(c) is from SAIL; the second row is 4SCALE.

TABLE III
FOUR-SCALE MODEL PARAMETERS

Parameter	Values
Component spectra	Leaf and Soil Spectral libraries
Diffuse sky light factor	0.1
structural parameters (GE_CHOICE, SHAPE, Ha(m), Hb(m), B(m2), D, m2, OMEGA_E)	NO_BRANCH, SPHEROID, 10.0, 7.0, 10000,1000, 2, 0.98
LAI	2.0, 4.0, 5.0, 6.0, 7.0
Illumination-view geometry	MODIS true observation geometry

Parameters: Ha—height of the lower part of the tree,Hb—height of cylinders, B-domain size,D—Number of trees in the domain,m2—cluster mean size, OMEGA_E—clumping index for shoots

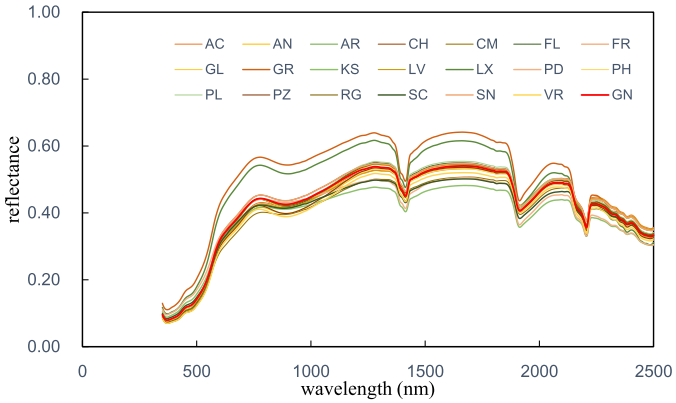


Fig. 3. Averaged soil profiles of each unit according to the FAO 1988 and the general soil unit (for abbreviations, refer to Table I)

and so on. Multi-sensor data are collected from Terra and Aqua/MODIS, Advanced Very High Resolution Radiometer/ National Oceanic Atmospheric Administration 18 (AVHRR/ NOAA18), and Visible and Infra-Red Radiometer/FengYun3 (VIRR/ FY3) from July to August in 2011. Preprocesses have been done to produce uniform BRF datasets including geometric co-registration and atmospheric correction. The *in situ* albedo is averaged from multiplot measurements over 3×3 km^2 homogeneous quadrates to obtain the reference true value for 1 km^2 pixel (sampling details please refer to [40]). Meanwhile, the corresponding components spectra are the typical spectra collected in the local area during the experiment in each experimental pixel.

V. PARAMETERIZATION RESULTS AND EVALUATIONS

This section presents how to derive and determine parameterized spectra. And then, the uncertainty and sensitivity of the new parameterization are examined to ensure its feasibility for global complexity.

A. Derivation of the Soil and Ratio Spectra

The soil spectra $\rho_g(\lambda)$ and the ratio spectra $\gamma(\lambda)$ are to be determined. Ideally, the soil spectra are assigned exactly according to the pixel location. However, it is not operational due to the efficiency and the limitation of a huge database. For ASK, the spectrum shape features along the wavelength are required but not their absolute values due to ASK coefficients' self-adjustment. Here, it is assigned according to the soil unit using the average profile to present their basic spectral shape in each unit. Fig. 3 shows the averaged profiles of each soil unit. In addition, a mean of all the averaged profiles has been defined as the general soil unit (abbreviated as GN) as a default unit. The GN can be taken as an average state of all the soil ignoring soil units. Shown in Fig. 3, the 20 averaged profiles have very similar features over the shortwave wavelength, although Unit GR and LX have larger values. This reveals the possibility of using just the general soil unit profile to stand for all the soil units, although it may introduce some uncertainty. This will be further investigated later.

Compared to the soil components, which can be categorized into some units, the leaf species is hard to be given for a pixel except a relatively homogenous pixel with a dominant vegetation species. And ASK is more sensitive to leaf than soil

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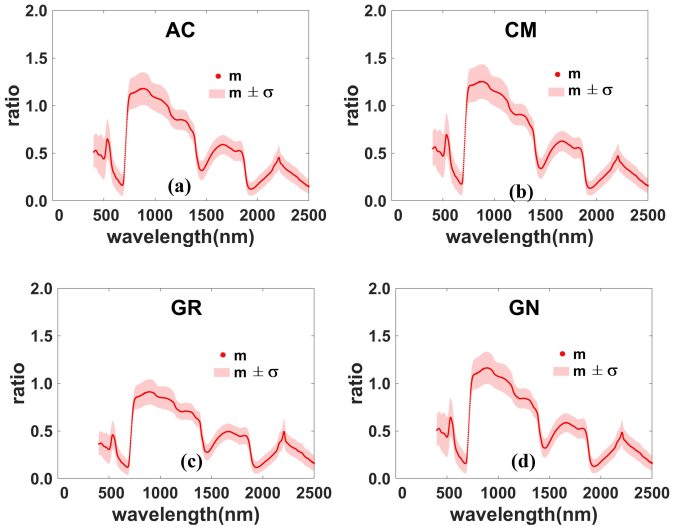


Fig. 4. Mean and standard variance of the ratio profiles for four soil unit examples. Ratio of (a) AC, (b) CH, (c) GR, and (d) GN (dot denotes the mean; patch denotes its variance within $\pm 1\sigma$).

from [30]. The ratio not only captures the differences between leaf and soil but also needs to quantify the leaf spectral variances. In this article, the ratio is statistically calculated from species of leaf spectra accompanied by each soil unit. The statistic of ratio is characterized by the form of mean (m) with levels (u) of standard deviation (σ), including m, $m \pm 0.5\sigma$, $m \pm 1\sigma$, to capture the average state and most of the boundary for a variable parameter. For convenience, this statistical parameter is denoted as $\gamma(m + u\sigma)$ eliminating the wavelength or spectral band. All the vegetation species are devoted to quantify the leaf variances. The mean and standard deviation are calculated for each unit including the 20 FAO units and the general soil unit.

Due to the similarity of the averaged soil profiles, the ratios of these units are similar along the wavelength. Fig. 4 presents some examples of the statistical ratio as the mean with one standard variance, i.e., $\gamma(m)$ and $\gamma(m \pm \sigma)$. The ratios' profiles have very similar shapes in the shortwave length domain. To quantify their difference, a mean relative difference is defined as (5) between two soil unit ratio profiles over the shortwave length range. The mean ratio $\gamma(m)$ has been evaluated here. Since the general soil (abbreviated as GN) presents the average state of all soil, it is taken as the benchmark, i.e., the difference is calculated between each FAO soil unit and the general soil unit. All the *RD*s of 20 FAO soil units are listed in Table IV. The *RD*s have inherited the soil reflectance spectrums' features in Fig. 3. As most of the FAO soil units' spectra have very closed values to the GN over the whole shortwave length, *RD*s are mostly around 3% revealing similar ratios as the GN, like AC and CM [which can also be visually compared from Fig.4(a) and Fig. 4(b)–(c)]. And, similar to soil spectrum profiles of GR and LX departures mostly from GNs, large *RD*s have appeared at these two soil units with 15.12% and 11.76%, respectively,

$$\mathrm{RD}_x = \mathrm{mean}\left(\frac{|\gamma_x(m,\Lambda) - \gamma_{GN}(m,\Lambda)|}{\gamma_{GN}(m,\Lambda)}\right) \tag{5}$$

TABLE IV
RATIO DIFFERENCES BETWEEN 20 FAO/UNESCO SOIL UNITS AND THE GENERAL SOIL UNIT

soil unit	Difference (%)	Soil unit	Difference (%)	Soil unit	Difference (%)
AC	2.20	GL	1.41	PZ	3.42
AN	3.31	GR	15.12	PD	6.92
AR	1.39	KS	9.62	RG	1.69
CM	6.11	LX	11.76	SC	4.89
CH	2.28	LV	2.64	SN	2.28
FR	2.51	PH	3.87	VR	5.88
FL	2.06	PL	2.48		

where RD_x is the mean ratio difference between x soil unit and general soil unit in the shortwave length domain. $\gamma_x(m,\Lambda)$ is the mean ratio of x soil unit, $\gamma_{GN}(m,\Lambda)$ is the mean ratio of general soil unit.

$\gamma(m + u\sigma)$ is a dynamic with variance of u to be determined during the inversion. Since the BRDF model's inverse function is hard to get, iterative techniques are routinely used to solve the kernel coefficients. The coefficients are iteratively adjusted until the modeled reflectance gets very close to the observed ones by minimizing the cost function [41], [42]. For the parameterized ASK, the cost function is rewritten as (5) eliminating the observation geometry and wavelength. It is a function of u with five discretized values (0, ± 0.5, ± 1) as candidates for each soil unit. The best solution is selected in the case of u with the minimum value among the five to-be-selected candidates. In the following description, we call this new ASK inversion mode as the optimum mode, compared to the original inversion with exactly true component spectra as input

$$\mathrm{Cost}(u) = \sum_{n=1}^{N} w_n \left(R'_n - R_n(\rho_g, \gamma(m + u\sigma))\right)^2 \tag{6}$$

where R'_n, $n = 1, \ldots, N$ is the nth observed reflectance, N is the total number of observations, R_n denotes the ASK simulated reflectance in the given geometry as the observation, and w_n is the weight with respect to each observation.

B. Performance Evaluation of the Parameterization

This parameterization methodology uses an averaged soil profile to represent the soil unit family features, and the ratios to specify the differences between leaf and soil component with standard deviation quantifying leaf variances. Comparing to the original inversion mode, it brings simplification at the cost of losing some accuracy. As the ratio is accompanied by the corresponding soil unit as one group of parameters, they will be evaluated in soil units. First, the effectiveness of using the exact soil unit parameters for ASK inversion will be tested, where the exact soil unit is the unit that the soil spectra used in BRF simulation. Then, the wrong soil unit will be assigned for inversion. Furthermore, the general soil unit is used by ignoring soil unit information or in the case of missing soil spectra. The three cases are compared with the original inversion in BRDF fitting and albedo estimation. For easy expression, we denote the original

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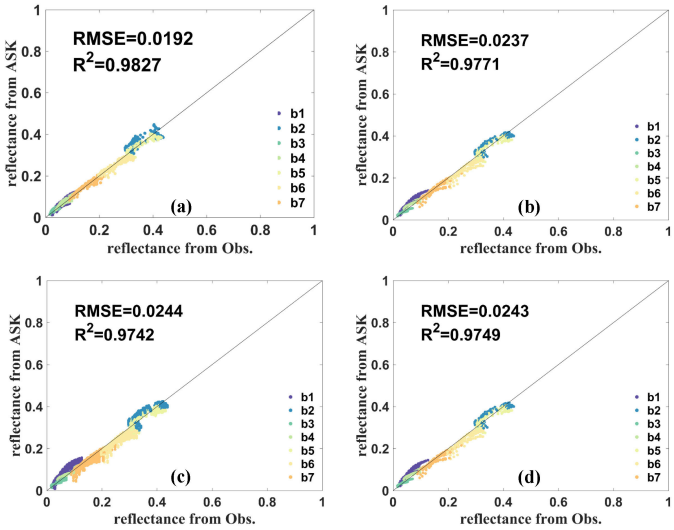


Fig. 5. BRF fitting comparisons. (a) True case. (b) Exact soil unit case. (c) Wrong soil unit case. (d) General soil unit case.

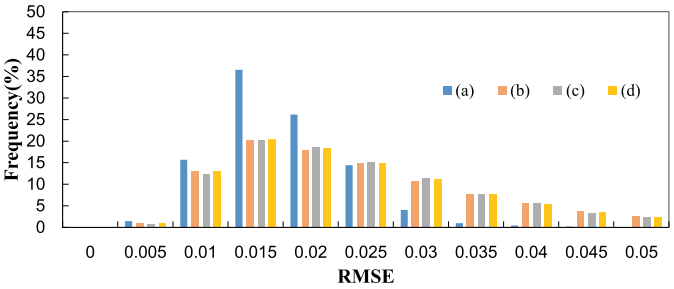


Fig. 6. BRF fitting RMSE distribution in all BRFs. (a) True case. (b) Exact soil unit case. (c) Wrong soil unit case. (d) General soil unit case.

TABLE V
MEAN OF THE BRDF FITTING RMSE FOR THE FOUR CASES AT DIFFERENT LAI SCENES AND ALL SCENES

Scenes	Input spectra			
	(a)true	(b)exact soil	(c)wrong soil	(d)general soil
Low LAI	0.0150	0.0306	0.0304	0.0297
Medium LAI	0.0185	0.0238	0.0237	0.0236
High LAI	0.0185	0.0218	0.0218	0.0217
All	0.0178	0.0248	0.0247	0.0245

inversion with true component spectra as the true case (case a), and the three cases used to evaluate the parameterization as the exact soil unit case (case b), wrong soil unit case (case c), and general soil unit case (case d), respectively.

In this evaluation, four angles, randomly chosen by combining seven bands to retrieve the coefficients in each BRF dataset, are enough to achieve full inversion when combining seven MODIS bands according to [25]. Then, the remaining 28 angles are used to validate the accuracy of forward simulation, and the shortwave Black Sky Albedo (BSA) and White Sky Albedo (WSA) are integrated from ASK by the retrieved coefficients and corresponding spectra input. For equivalent comparisons, the angular sampling combination is the same in each BRF dataset, while the input spectra are different during the inversions.

1) Effects in the BRDF Fitting: Generally, all the determined R^2 coefficients with values above 0.95 indicate good linear consistency between the ASK forward simulation and observations. Therefore, the following analysis will focus on the root-mean-square error (RMSE).

Fig. 5 presents BRDF fitting examples in the four cases. The ratio with the exact unit soil spectra reveals a slightly poorer fitting with an RMSE of 0.0237 compared to the true case with an RMSE of 0.019. An acceptable accuracy is reached for the ratio with exact soil unit spectra. While the wrong unit of soil is assigned as shown in Fig. 5(c), similar fitting accuracy appears. Their RMSE shows that the exact soil unit performs slightly better than the wrong soil unit. Note that the ratio with the wrong soil unit evaluations are done with all the other 19 soil units, respectively, except the exact soil unit to which the BRFs observations belong to. Therefore, more scatters are plotted in Fig. 5(c) than the other subfigures. For some wrong soil units, they present better fitting (or worse) as some scatters are a bit close to (or discrepant from) the 1:1 line. However, the average accuracy is comparable to the accuracy of the exact soil unit within 0.001 RMSE differences. In some degrees, case (c) can be taken as an average state of all soil spectra, which exhibits a promising possibility of using the general soil unit instead of the exact soil unit. As shown in Fig. 5(d), the general soil unit also presents an RMSE which is very close to that of the exact soil unit case.

Fig. 6 presents the RSME distribution of all the BRF datasets, and Table V summarizes their statistics. Compared to the true case, the fitting results of the new parameterization cases appear slightly poorer, where their RMSE frequencies in the range from 0.015 to 0.020 are about 10% to 15% less and 3% to 7% more among the range of 0.025 to 0.05, which are shown as series of (b)–(d) compared to series (a) in Fig. 6. The wrong soil unit case (c) and the general soil unit case (d) have very similar distribution as the exact soil unit case (b), which indicates that the new inversion is not very sensible to the soil unit. The statistics show the same results as the mean of these RMSEs in the three cases are nearly the same as 0.0248, 0.0247, and 0.0245, respectively, as shown in Table V. The general soil unit inversion presents an acceptable loss of accuracy of about 0.0067 (mean RMSE is 0.0245 slightly higher than the true case of 0.0178). Thus, we can use the general soil unit to represent all the soil units.

Such small accuracy loss in the absence of accurate soil information using the general soil unit can be ascribed into two aspects. One is due to the similar reflectance features of these soil units in the shortwave length domain as shown in Fig. 3. The ratios of these soil units calculated from their reflectance have then inherited such similarity (shown in Fig. 4), where most of their mean relative differences are around 3% compared to the general soil unit (listed in Table IV). The other one is the new inversion strategy of selecting $\gamma\,(m + u\sigma)$, which permits the mathematically best fitting in the semiempirical model. u can have a compensation with soil spectra to adjust the ASK coefficients. When the spectral parameter input has some degrees of biases from

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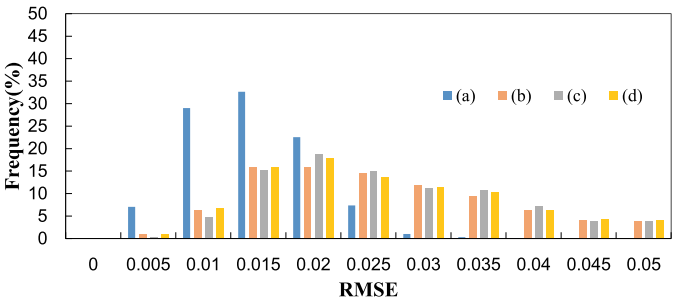


Fig. 7. BRF fitting RMSE distribution in low LAI scenes. (a) True case. (b) Exact soil unit case. (c) Wrong soil unit case. (d) General soil unit case.

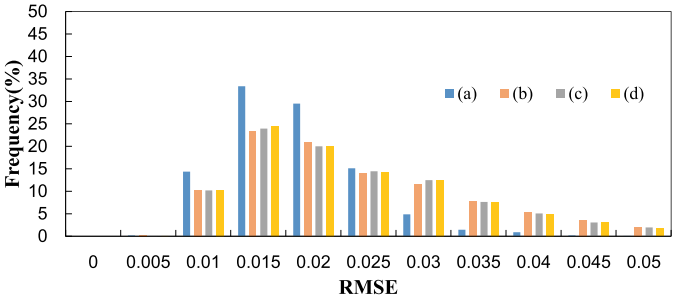


Fig. 8. BRF fitting RMSE distribution in medium LAI scenes. (a) True case. (b) Exact soil unit case. (c) Wrong soil unit case. (d) General soil unit case.

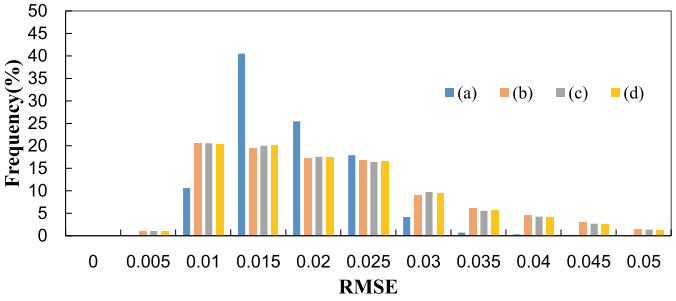


Fig. 9. BRF fitting RMSE distribution in high LAI scenes. (a) True case. (b) Exact soil unit case. (c) Wrong soil unit case. (d) General soil unit case.

the exact parameters, the coefficients can self-adjust with the semiempirical nature to minimize the total residuals for best BRDF fitting during the retrieval. This mathematical fitting ability allows more tolerance for the loss of accurate soil information. With these two factors, the general soil unit can satisfy the inversion accuracy while not using the exact soil unit.

Although it is hard to differentiate the roles of the soil and the ratio as they are treated as a whole in the new parameterization, different LAIs are separately counted to assess the parameterization's performance in detail, as shown in Figs. 7–9. In low LAI scenes, the BRF spectra are dominated by the soil component. We can evaluate the feasibility of the parameterized soil. As shown in Fig. 7, case (a) yields better fitting as more distributed in the range of (0–0.02) than the other three cases (b)–(d). Their mean RMSEs are about 0.015 higher than the true case listed in Low LAI scenes in Table V. For the medium LAI scenes, the results are very similar to all LAI scenes, shown in Fig. 8 by comparing to Fig. 6. Their precisions are a little lower than the true case with a loss of about 0.005 in RMSE. For high LAI scenes, the BRFs present more vegetation features, which will be more sensible to the leaf component. As shown in Fig. 9 and according to the mean RMSE of High LAI scenes shown in Table V, better retrieval results are presented with about 0.003 precision loss compared to the true case. As the optimum mode involves dynamic leaf variances in u to minimize the cost function, it enlarges the possibility of better mathematical fitting when the scenes present more vegetation features, which reveals that the ratio with soil spectra has well captured the spectral difference in ASK and achieves a satisfactory accuracy.

Although true case (a) has better BRF fitting than the optimum mode (b, c, d), in general, for high LAI scenes, the optimum mode inversions (b, c, d) may perform better than the true case in some situations. As shown in Fig. 9, higher frequency appears in small RMSE (ranging from 0.005 to 0.01). This may be caused by the vegetation component spectra input in some densely vegetated scenes. The canopy reflectance is required in the geo-kernel that is constructed by the sunlit canopy and sunlit soil ground to simulate the soil-vegetation-composited scenes. The true case (a) has input leaf reflectance as leaf reflectance is easily achieved in practice while canopy reflectance is hard to be measured or calculated from leaf reflectance because of the variable canopy structures. However, canopy reflectance is different from leaf reflectance. The true case (a) still has some unavoidable errors even using the true leaf reflectance, especially in some densely vegetated scenes. While for the optimum mode (b,c,d), the ratio has involved five candidates u (0, ± 0.5, ± 1) of standard deviation (σ) for best BRF fitting, which has captured the boundary of the variable vegetation reflectance. In some situations, such new parameterization can partially play the role of canopy reflectance approximated from leaf reflectance to achieve a better BRF fitting.

2) Effects in the Albedo Estimation: Albedo is a further evaluation parameter for BRDF model retrieval. The retrieved coefficients and corresponding spectra are utilized to integrate the shortwave BSA and WSA, where shortwave BSA is derived at a sun zenith angle of 30°. Fig. 10 compares the three new parameterization cases (i.e., b, c, and d) with the true case (a). Note that, here, the true case (a) is also derived from BRDF model integration, which is taken as a reference but not a true value.

Since albedo is the integration of BRDF over angular and spectral domains, all the three cases present similar albedo estimations inheriting the BRF fitting patterns. The scatters are distributed similarly in the three subfigures of the first row shown in Fig. 10(a) and (c). In Fig. 10(b), of using the wrong soil unit, some scatters locate nearer to the line of 1:1, and some a bit far apart from 1:1 line. And an average state has been achieved as the exact soil case. This again proves the applicability of using the general soil unit to represent all soil units. WSA has similar situations as BSA. The RMSE is about 0.0116 for BSA as shown in Fig. 10(c), and 0.0182 for WSA in Fig. 10(f). The biases between the general soil unit case and the true case are acceptable taking the albedo application precision requirement of 0.05 as a benchmark [43].

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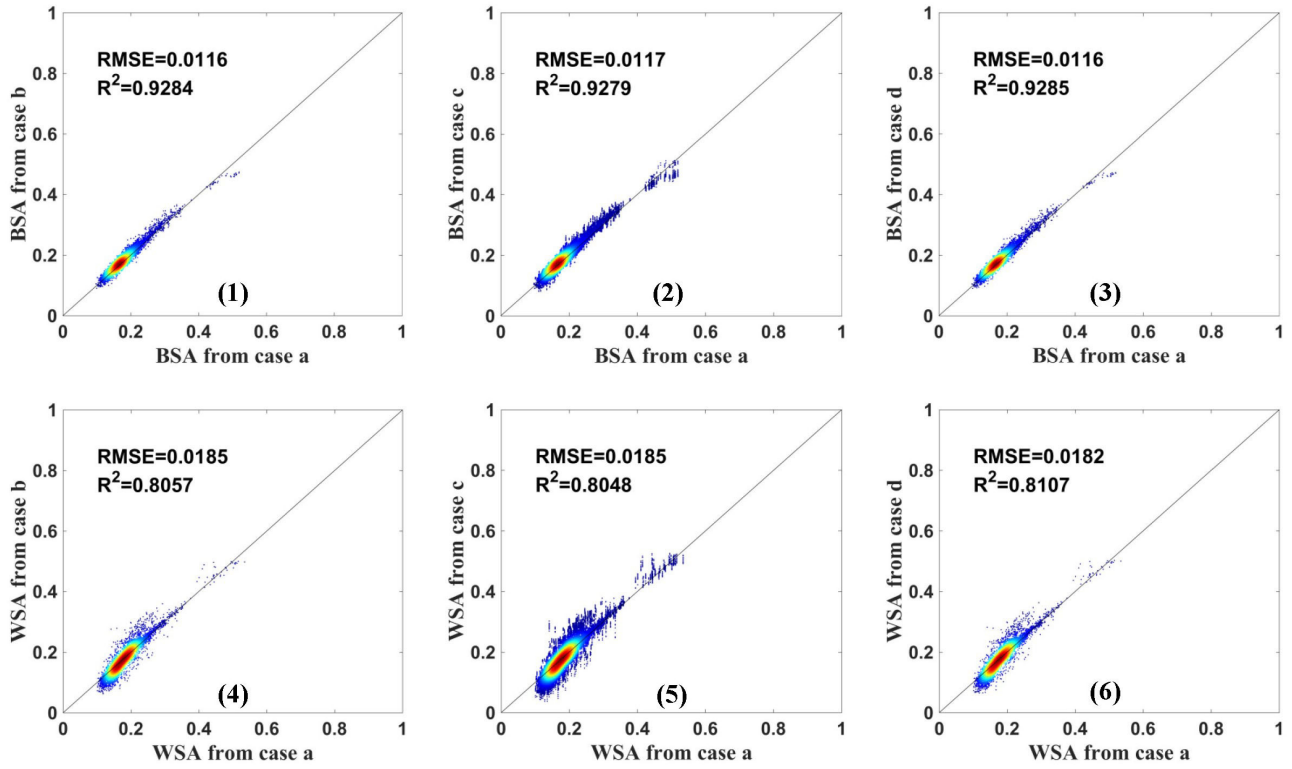


Fig. 10. Albedo estimations from different cases at a solar zenith angle of 30° for (a)–(c) BSA and for (d)–(f) WSA, with the exact soil unit (a) and (d) case b, (b) and (e) wrong soil unit case c, and (c) and (f) general soil unit case d compared to the true case (a).

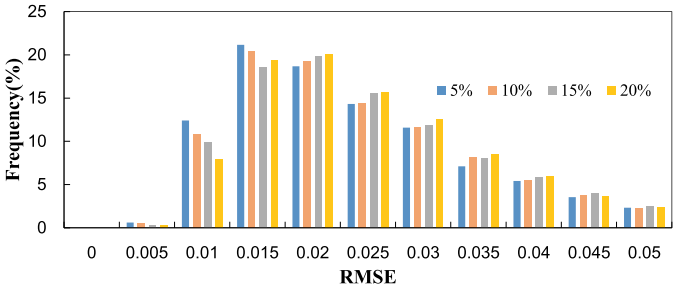


Fig. 11. RMSE distribution of BRF fitting with soil noises at levels of 5%, 10%, 15%, and 20%.

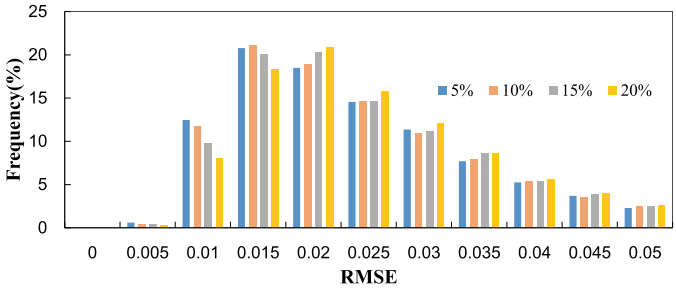


Fig. 12. RMSE distribution of BRF fitting with leaf noises at levels of 5%, 10%, 15%, and 20%.

C. Robustness Assessment of the Parameterization

Shown as the primary evaluation in Section V-B, the general soil and its corresponding ratio have presented promising potentials. Nevertheless, further robustness evaluation is required to assure the feasibility of using ratio with the general soil unit for global complicated situations.

Random sources of noises (5%, 10%, 15%, and 20% normally distributed random noises) are added in the spectral profiles as shown in (7), accounting for the spectra disturbance caused by the temporal and spatial variance of leaf or soil and the spectral samples' limitation in the parameterization process. Three specified situations are discussed as: 1) noise sourced from the general soil; 2) noise sourced from the statistical ratio; and 3) noise from the two at the same time. Four angles are randomly chosen for inversion. Then, the coefficients and the corresponding noised spectra are used

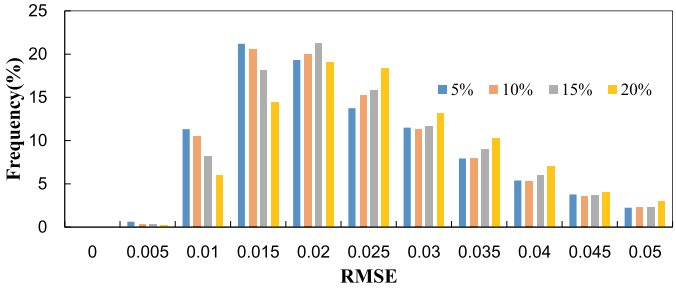


Fig. 13. RMSE distribution of BRF fitting with noises in both in soil and ratio at levels of 5%, 10%, 15%, and 20%.

to forward construct the remaining BRFs and integrate the shortwave BSA and WSA

$$r_e(\lambda) = r(\lambda) + r(\lambda) \cdot P \cdot E(\lambda) \tag{7}$$

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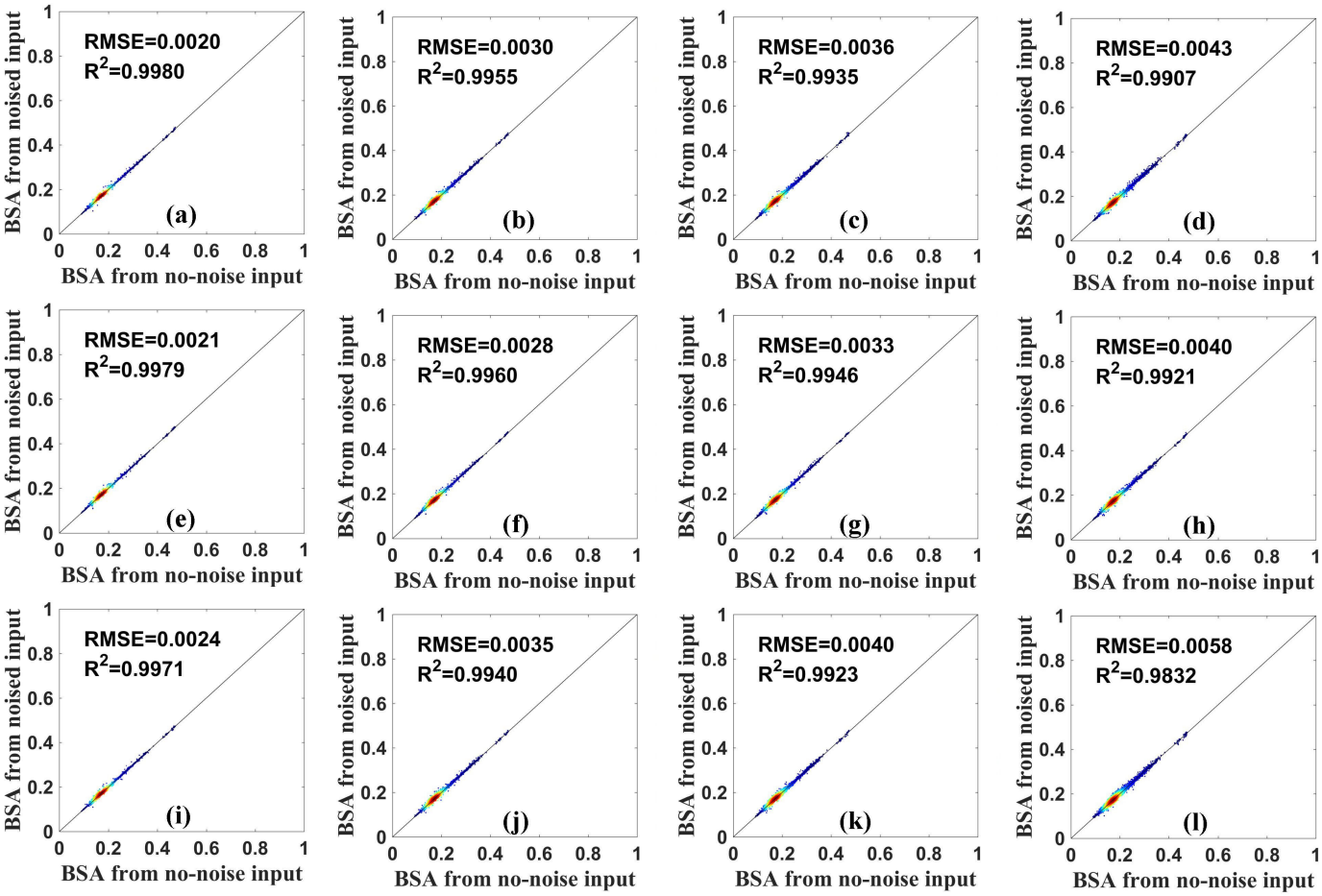


Fig. 14. BSA comparisons between the different noises and the nonnoised derivation. (a)–(d) Noises soil at the level of 5%, 10%, 15%, 20%, respectively. (e)–(h) Noised ratio from 5%–20%. (i)–(l) Noised soil and ratio from 5% to 20%.

TABLE VI
MEAN OF THE BRF FITTING RMSES

Error source	Error level			
	5%	10%	15%	20%
Soil	0.0245	0.0250	0.0256	0.0259
Ratio	0.0246	0.0247	0.0252	0.0260
Soil & ratio	0.0248	0.0250	0.0259	0.0277

TABLE VII
BSA RMSE

Error source	Error level			
	5%	10%	15%	20%
Soil	0.0020	0.0030	0.0036	0.0043
Ratio	0.0021	0.0028	0.0033	0.0040
Soil & ratio	0.0024	0.0035	0.0040	0.0058

TABLE VIII
WSA RMSE

Error source	Error level			
	5%	10%	15%	20%
Soil	0.0027	0.0039	0.0046	0.0057
Ratio	0.0025	0.0036	0.0043	0.0050
Soil & ratio	0.0032	0.0045	0.0052	0.0075

where $r_e(\lambda)$ is the parameterized spectra with uncertainty, $r(\lambda)$ is the accurate spectra, and the noise is calculated as the accurate value $\gamma(\lambda)$ of a noise level P (5%, 10%, 15%, 20%), and random value $E(\lambda)$ which are generated as a normal distribution with a std$(E) = 1$ and mean$(E) = 0$.

1) Assessments in BRF Fitting: Figs. 11–13 present the RMSE distributions of BRF fitting when the parameterized spectra are noised by different sources and levels of noises, and Table VI gives their mean RMSE.

First, we focus on the different levels' effects within each source of noise; 5% random error in all the three sources brings a negligible bias from the no-noise retrieval which is shown as the RMSEs distribution and the average of RMSEs (comparing the 5% noise column in Table VI as 0.0245, 0.0246, 0.0248 to 0.0245). As shown in Figs. 11–13, the proportion of RMSEs with low values around 0.05–0.15 has decreased with the increasing magnitude of noise. However, only about 2% decrement happens with every 5% noise addition. And, an increase of less than 0.001 in the average RMSEs occurs with every 5% noise increment, except the extreme precision loss of 0.002 occurred for 20% noise compared to 15% added in soil and ratio. Such slight precision loss reveals stable retrieval when noises within 20% in each noise source.

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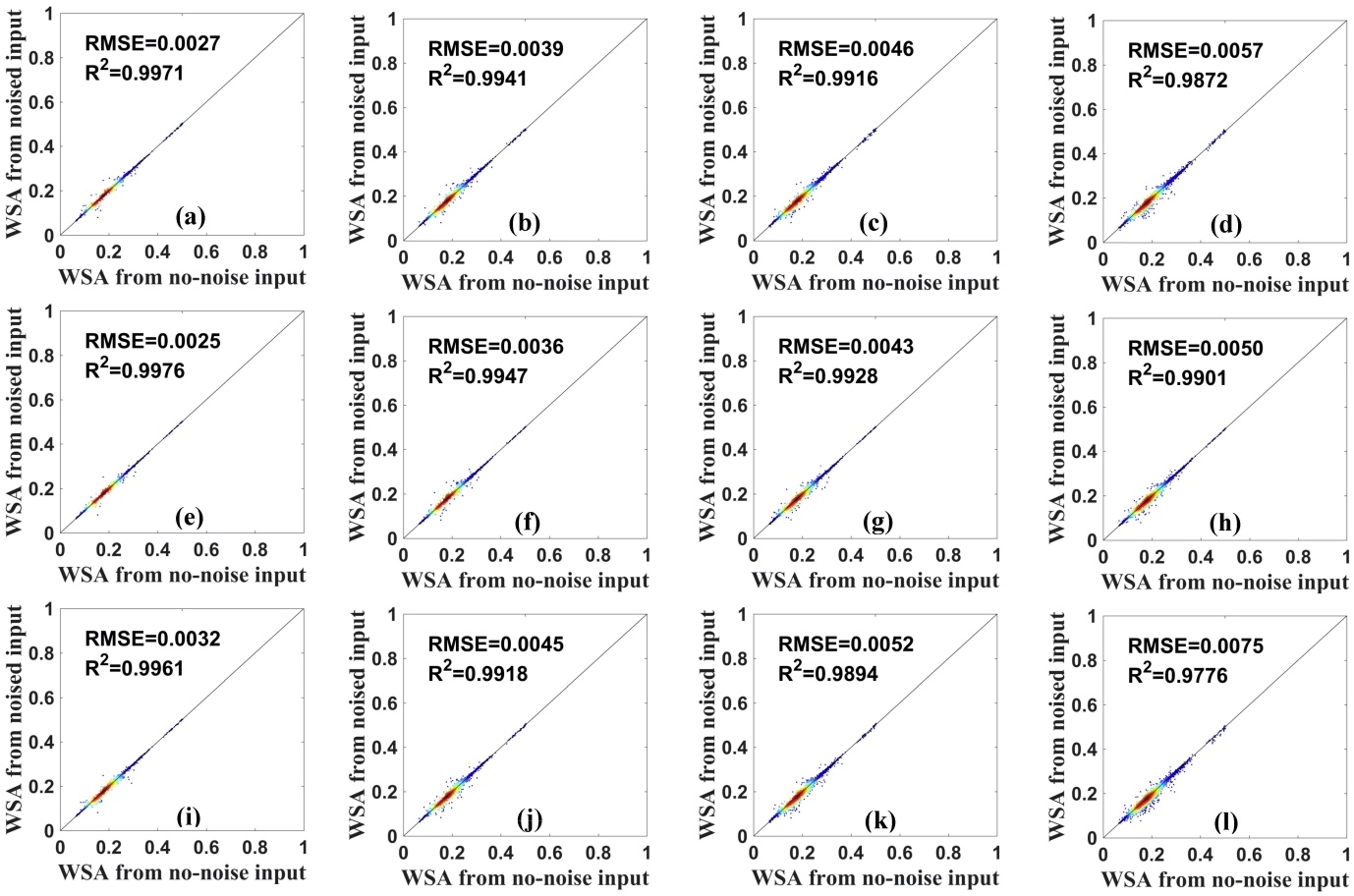


Fig. 15. WSA comparisons between the different noises and the nonnoised derivation. (a)–(d) Noises soil at the level of 5%, 10%, 15%, and 20%, respectively. (e)–(h) Noised ratio from 5%–20%. (i)–(l) Noised soil and ratio from 5% to 20%.

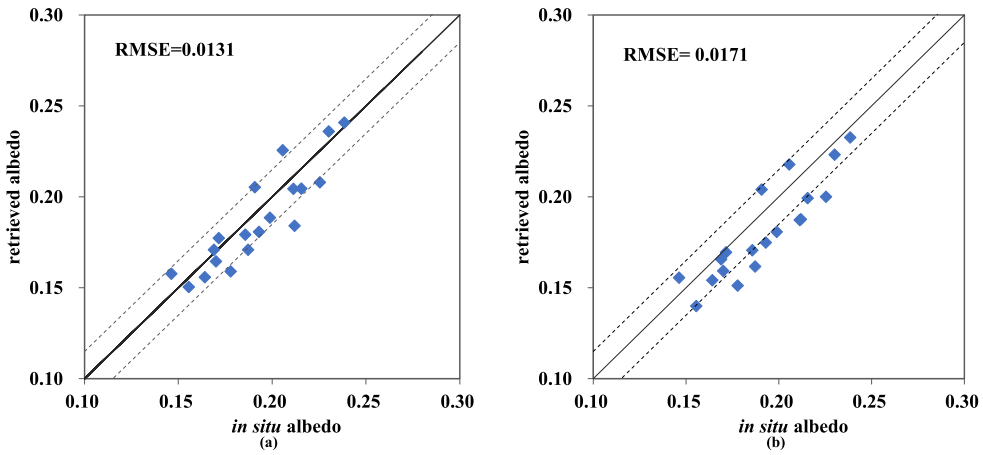


Fig. 16. Validation of satellite albedo retrievals from MODIS, AVHRR, and VIRR in a four-day temporal resolution. (a) Original mode with true component spectra input. (b) Optimum mode with the general soil unit and ratio. The two dashed lines in each panel show the bias range between [−0.015, 0.015].

Second, we investigate the different error sources. They exhibit no significant differences as listed in Table VI. The single source of noises from soil or ratio conducted nearly equivalent and ignorable effects in the retrieval as the similar distribution of RMSEs shown in Figs. 11 and 12 causing 0.001 to 0.004 differences in average RMSEs at the same level noise. Even when noised by soil and ratio (Fig. 13), insignificant effects have been caused with the largest biases of 0.0017–0.0018 in the level of 20% noise shown in the last column of Table VI. Hence, the noise effects from soil and ratio are not a simple accumulating influence in the retrieval. This proves the compensation between the soil and ratio during the inversion.

Regarding the RMSE fitting accuracy, these noises have resulted in a little precision loss. The most noised retrievals (i.e., 20% noise added to soil and ratio) has a mean fitting

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TABLE IX
VALIDATION OF RETRIEVED ALBEDO

Date	Site name	Lon/Lat	in situ	Original mode	Optimum mode
20110725	JC Barrens	100.70°E/ 38.75°N	0.1555	0.1504	0.1400
20110729	HZZ Barrens	100.32°E/ 38.76°N	0.1872	0.1709	0.1617
20110808	HZZ Barrens	100.32°E/ 38.76°N	0.1857	0.1792	0.1706
20110809	BJ Barrens	100.26°E/ 38.93°N	0.2254	0.2080	0.2000
20110811	HPX Barrens	100.47°E/ 38.72°N	0.1988	0.1886	0.1807
20110821	JC Barrens	100.70°E/ 38.75°N	0.1462	0.1577	0.1555
20110822	JC Barrens	100.70°E/ 38.77°N	0.1641	0.1559	0.1542
20110824	HZZ Barrens	100.32°E/ 38.76°N	0.1689	0.1709	0.1657
20110825	HPX Barrens	100.47°E/ 38.71°N	0.2155	0.2046	0.1993
20110827	HZZ Barrens	100.32°E/ 38.76°N	0.1930	0.1807	0.1749
20110828	JC Barrens	100.70°E/ 38.75°N	0.1700	0.1645	0.1593
20110802	EG Gobi	100.91°E/ 42.07°N	0.2056	0.2256	0.2178
20110803	EG Gobi	101.15°E/ 42.38°N	0.2113	0.2044	0.1873
20110806	HC Gobi	100.92°E/ 41.61°N	0.1907	0.2053	0.2040
20110726	BDK Rape flower	101.10°E/ 38.22°N	0.2301	0.2360	0.2232
20110822	BDK Rape flower	101.08°E/ 38.26°N	0.2119	0.1841	0.1877
20110824	YK Corn	100.41°E/ 38.86°N	0.1778	0.1590	0.1512
20110731	Dz Desert	100.56°E/ 38.50°N	0.1715	0.1773	0.1695
20110805	EG Saline alkali soil	100.71°E/ 42.21°N	0.2385	0.2409	0.2327

RMSE of 0.0277. A precision loss of 0.0032 occurred when compared to 0.0245 with no noise of the general soil unit. Even compared to the true case with a mean RMSE of 0.0178, the precision has a loss of 0.0099. Hence, a good robustness is shown to the spectral noises.

2) Assessments in Albedo Estimation: Fig. 14 compares the shortwave BSA from differently noised parameterizations to the nonnoised ones, and Table VII summarizes the RMSE in different noise cases. Fig. 15 is the WSA, and their RMSE are summarized in Table VIII. As the WSA has similar results as BSA with about within 0.002 higher RMSE compared to the BSA counterparts, we will focus on the analysis of BSA. All subfigures show good consistency between the compared pairs with high R^2 of 0.98–0.99. Similar trends inherit from the BRF fitting. At a low level of noise as 5%–10% in the first two columns of Fig. 14, the three sources have little differences as nearly the same RMSE around 0.002, 0.003 listed in Table VII. When increased to 20% as shown in the last column of Table VII, some obvious uncertainties appear as RMSE is 0.0043 for the noised soil in Fig. 14(d), 0.0040 for the ratio in Fig. 14(h), and 0.0058 for noise in soil and ratio. However, the differences are acceptable compared to the precision requirement within 0.05 in the albedo application.

The random noises evaluation proves good robustness of the parameterization to the spectra uncertainty. It may be identified by three inherent properties of the new parameterization. First, the parameterization is a statistics from various species of leaves and types of soil, where statistical quantities are not very sensitive to noise. Second, the ratio is designed as an indivisible whole of leaf and soil. A compensation between soil and ratio makes no significant differences in the different noise sources and ensures an acceptable solution while minimizing the cost function. Third, the noise is added on each wavelength, while sensors' observations are an integration response over a narrow waveband. In this way, the noises may be smoothed and not so sound in the narrow band.

VI. FEASIBILITY ASSESSMENT OF SATELLITE DATA

Synergistic retrieval of albedo combining multisensor data in the last article presents promising potential if the exactly true component spectra are input. Whether the parameterized ASK can still satisfy the accuracy requirement needs to be assessed. Same processes of combining multisensor data for retrieval are done here to examine the new parameterization feasibility. The BRDF is retrieved in a four-day composited temporal resolution combining clear-sky observations, and then shortwave BSA and WSA are integrated. Shortwave blue-sky albedo for direct comparisons with *in situ* albedo is interpolated between BSA and WSA with the diffuse skylight factor simulated from the 6S (second simulation of the satellite signal in the solar spectrum) model with field-measured aerosol optical depth and water vapor input.

Fig. 16 compares the satellite retrievals to the *in situ* albedo, and the detailed information is listed in Table IX. (a) is the original mode with true component spectra from [26] and (b) is the newly proposed optimum mode of the general soil unit. The accurate input of component spectra (i.e., for each land cover type, the components' spectra are the typical spectra collected locally) presents a high precision of albedo derivation with a RMSE of 0.0131. However, it is not operational for a

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global inversion due to the efficiency considering the spatial and temporal variability of component spectra. In addition, it is hard to define the exact component spectra, especially for a mixture pixel. The parameterized spectra, although with less precision compared to the true component spectra counterpart, can still achieve good agreement with *in situ* measurements with a higher RMSE of 0.0171.

These results indicate satisfactory feasibility of the parameterized ASK for various scenes as barrens, Gobi, desert, rape flower, corn, and saline-alkali soil composed of different soil and leaf components. The parameterization, involving a large amount of leaf and soil spectra, can generally capture the leaf and soil spectral characteristics. In addition, five candidates (mean with five different levels of stand deviation) allow more tolerance for spectral uncertainty, and the optimum inversion takes the advantage of ASK self-adjustment from the semiempirical nature to make inversion more robust to the spectra uncertainty. Such two features in this parameterization assure the feasibility of global usage.

VII. Conclusion

This article proposed a simple parameterization involving global soil and leaf spectral libraries from ANGERS, GOSPEL, LOPEX, and USGS to parameterize the leaf and soil component spectra in the ASK model. The ratio of leaf spectra to soil spectra is calculated to specify the differences between them with five levels of standard deviation quantifying leaf spectral variances. Soil spectral profiles are averaged to capture the soil-unit-family features. In this way, the soil and corresponding ratio are a whole during the retrieval. An optimum retrieval strategy is employed to determine the ratio candidates, thus minimizing the cost function.

Investigation of the parameterization performance shows good effectiveness of using the general soil unit (calculated from all the soil samples ignoring soil units) with its corresponding ratio to retrieve BRDF and derive albedo. No significant difference happens when using the wrong soil unit instead of the exact soil unit. Compared to the original mode, the general soil unit with ratio conducts a precision loss of about 0.007 in BRF fitting, with a mean RMSE of 0.0245 slightly higher than the true component spectra inversion of 0.0178. The parameterization sacrifices some degree of information using just one general soil when scenes are dominated by soil, with a precision loss of about 0.015 higher RMSE in low LAI scenes. However, the candidates of five levels of standard derivation (0, ± 0.5, ± 1) in ratio allows more tolerance for spectral uncertainty and the optimum inversion takes advantage of the semiempirical nature of ASK to achieve a better inversion. Such an advantage has been explored in the medium to high LAI scenes with just 0.003–0.005 higher RMSE. For the albedo estimation, it also presents no significant difference compared to the true spectra retrieval with an RMSE of 0.0116 in shortwave BSA, and 0.0182 for WSA.

The optimum retrieval with general soil unit and corresponding ratio presents good robustness to noise. Noises either added into soil or ratio affected the retrievals insignificantly from 5% to 20% levels. Comparing the separated noise sourced from soil or ratio to the noise from both, no obvious accumulation effect proves the compensation between soil and ratio during inversions. Even when noise is added up to 20% in the soil and ratio, the BRF fitting RMSE is 0.0277 and BSA RMSE is 0.0058.

Application in the satellite data has proven the parameterization's practical feasibility. The synergistic retrieval of albedo in the Heihe River Basin combines MODIS/Terra+Aqua, AVHRR/NOAA8, and VIRR/FY3 in a four-day composited temporal resolution in different surface land cover types. The estimated albedo yields a satisfactory consistency with the *in situ* measurements as RMSE is 0.0171, comparable to the true component retrieval with an RMSE of 0.0131.

The new parameterized ASK model is promising for easy operation in the global application taking the advantages of the statistical spectral quantities from a large number of spectral samples and the semiphysical nature of the kernel-driven model. More validation in the real satellite data retrieval is required in the future work. While this article focused on the soil-vegetation system, a snow-ice kernel with corresponding spectra parameterization is a further improvement for the snow-ice-covered areas.

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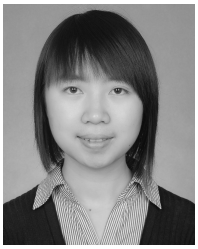
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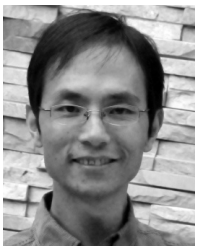
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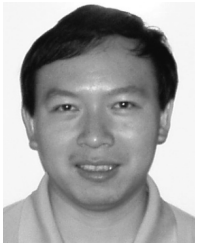
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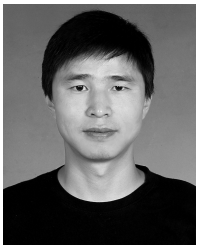
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